C16 Organic Analysis – Knowledge Organiser

AQA A-level Chemistry • Set 1: Mass Spectrometry and IR **Paper 2**

#	Question	Answer
1	Decimal places given by high resolution MS	4 or more
2	Use of high resolution MS	Distinguish compounds with the same integer M_r
3	Molecular ion peak	M^+, highest *m/z*
4	What absorbs infrared radiation	Vibrating bonds
5	x-axis of an IR spectrum	Wavenumber (cm^{-1})
6	y-axis of an IR spectrum	Transmittance (%)
7	Fingerprint region	Below 1500 cm^{-1}
8	Use of the fingerprint region	Identify a compound by comparison
9	Three greenhouse gases	CO_2, CH_4, H_2O
10	Radiation absorbed by greenhouse gases	Infrared

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AQA A-level Chemistry • Set 2: IR Absorptions **Paper 2**

#	Question	Answer
1	O–H (alcohol)	3230–3550 cm^{-1}, broad
2	O–H (carboxylic acid)	2500–3000 cm^{-1}, very broad
3	N–H (amine)	3300–3500 cm^{-1}
4	C–H	2850–3300 cm^{-1}
5	C≡N	2220–2260 cm^{-1}
6	C=O	1680–1750 cm^{-1}
7	C=C	1620–1680 cm^{-1}
8	C–O	1000–1300 cm^{-1}
9	Absorptions for an aldehyde or ketone	C=O only (no O–H)
10	Absorptions for a carboxylic acid	Very broad O–H and C=O